```
> restart:with(DifferentialGeometry):with(JetCalculus):with(DEtools)
  :with(PDEtools):with(IntegrationTools):
> DGsetup([t,x],[u,v,w],E,10):
E >
E > ##### STEP 1
E >
E > ### component form of the elastic string PDEs
E >
E > strain:=lambda=sqrt( (1+u[2])^2+v[2]^2+w[2]^2)-1;
```

$$strain := \lambda = \sqrt{\left(1+u_2\right)^2 + v_2^2 + w_2^2} - 1 \tag{1}$$

```
E >
E > # express k(lambda) as solution of differential equations
E > declare(k(u[2],v[2],w[2]));
```

$$k\left(u_2, v_2, w_2\right) \textit{ will now be displayed as } k \tag{2}$$

```
E > k_eqns:=[diff(k(u[2],v[2],w[2]),u[2])/(1+u[2])=diff(k(u[2],v[2],w
  [2]),v[2])/v[2],diff(k(u[2],v[2],w[2]),u[2])/(1+u[2])=diff(k(u
  [2],v[2],w[2]),w[2])/w[2],diff(k(u[2],v[2],w[2]),w[2])/w[2]=diff
  (k(u[2],v[2],w[2]),v[2])/v[2]];
```

$$k\_eqns := \left[\frac{k_{u_2}}{1+u_2} = \frac{k_{v_2}}{v_2}, \frac{k_{u_2}}{1+u_2} = \frac{k_{w_2}}{w_2}, \frac{k_{w_2}}{w_2} = \frac{k_{v_2}}{v_2}\right] \tag{3}$$

```
E > pdsolve(k_eqns):subs({_F1(2*u[2]+u[2]^2 + v[2]^2 + w[2]^2)=k
  (lambda)},%[]):subs(strain,%):k_funct:=%;
```

$$k\_funct := k = k\left(\sqrt{\left(1+u_2\right)^2 + v_2^2 + w_2^2} - 1\right) \tag{4}$$

```
E >
E >
E > eqn_u:=u[1,1]-TotalDiff(k(u[2],v[2],w[2])*(1+u[2]),[2]);
```

$$eqn\_u := u_{1,1} - \left(k_{u_2}\left(1+u_2\right) + k\right) u_{2,2} - k_{v_2}\left(1+u_2\right) v_{2,2} - k_{w_2}\left(1+u_2\right) w_{2,2} \tag{5}$$

```
E > eqn_v:=v[1,1]-TotalDiff(k(u[2],v[2],w[2])*v[2],[2]);
```

$$eqn\_v := v_{1,1} - k_{u_2} v_2 u_{2,2} - \left(k_{v_2} v_2 + k\right) v_{2,2} - k_{w_2} v_2 w_{2,2} \tag{6}$$

```
E > eqn_w:=w[1,1]-TotalDiff(k(u[2],v[2],w[2])*w[2],[2]);
```

$$eqn\_w := w_{1,1} - k_{u_2} w_2 u_{2,2} - k_{v_2} w_2 v_{2,2} - \left(k_{w_2} w_2 + k\right) w_{2,2} \tag{7}$$

```
E >
E > pdes:=[eqn_u,eqn_v,eqn_w]:
E > solve(pdes,[u[1,1],v[1,1],w[1,1]])[]:lead_ders:=%:
E > ord:=2;
```

$$ord := 2 \tag{8}$$

```
E >
E >
```

```
E >
E > ### component form of point symmetry
E >
E > dep:=t,x,u[],v[],w[]:
E > P_u:=eta_u(dep)-xi(dep)*u[2]-tau(dep)*u[1];
```

$$P\_u := eta\_u(t, x, u[\,], v[\,], w[\,]) - \xi(t, x, u[\,], v[\,], w[\,])\, u_2 - \tau(t, x, u[\,], v[\,], w[\,])\, u_1 \quad (9)$$

```
E > P_v:=eta_v(dep)-xi(dep)*v[2]-tau(dep)*v[1];
```

$$P\_v := eta\_v(t, x, u[\,], v[\,], w[\,]) - \xi(t, x, u[\,], v[\,], w[\,])\, v_2 - \tau(t, x, u[\,], v[\,], w[\,])\, v_1 \quad (10)$$

```
E > P_w:=eta_w(dep)-xi(dep)*w[2]-tau(dep)*w[1];
```

$$P\_w := eta\_w(t, x, u[\,], v[\,], w[\,]) - \xi(t, x, u[\,], v[\,], w[\,])\, w_2 - \tau(t, x, u[\,], v[\,], w[\,])\, w_1 \quad (11)$$

```
E >
E > symm_vars:=[indets({P_u,P_v,P_w},function)[]]:declare(%):
```

$eta\_u(t, x, u[\,], v[\,], w[\,])$ *will now be displayed as* eta_u

$eta\_v(t, x, u[\,], v[\,], w[\,])$ *will now be displayed as* eta_v

$eta\_w(t, x, u[\,], v[\,], w[\,])$ *will now be displayed as* eta_w

$\tau(t, x, u[\,], v[\,], w[\,])$ *will now be displayed as* τ

$\xi(t, x, u[\,], v[\,], w[\,])$ *will now be displayed as* ξ **(12)**

```
E >
E > symm_generator:=P_u*D_u[]+P_v*D_v[]+P_w*D_w[]:
E >
E >
E >
E > ##### STEP 2
E >
E > ### determining equations
E >
E >
E > # apply symmetry generator to each pde
E > symm_pdes:=[seq(Hook(Prolong(symm_generator,ord),
    VerticalExteriorDerivative(pdes[i])),i=1..nops(pdes))]:indets(%,
    name):select(has,%,1):{seq([op(%[i])],i=1..nops(%))}:select(has,
    %,{[1,1],[1,1,1],[1,1,2]});
```

$$\{[1, 1], [1, 1, 1], [1, 1, 2]\} \quad (13)$$

```
E >
E > # substitute u_tt,v_tt,w_tt, and their t,x derivatives
E > subs(TotalDiff(lead_ders,[1]),symm_pdes):subs(TotalDiff
    (lead_ders,[2]),%):subs(lead_ders,%):deteqns:=simplify(%):nops(%)
    ;
```

$$3 \quad (14)$$

```
E > indets(deteqns,name):select(has,%,1):{seq([op(%[i])],i=1..nops(%)
```

```
)} intersect {[1,1],[1,1,1],[1,1,2]};
```

$$\varnothing \tag{15}$$

```
E >
E > # split in jet space (derivatives of u,v,w) excluding (u_x,v_x,
    w_x) in k
E > split_vars:={(indets({deteqns},name) minus indets({symm_vars,
    strain},name))[]};
```

$$split\_vars := \{u_1, u_{1,2}, u_{2,2}, u_{1,2,2}, u_{2,2,2}, v_1, v_{1,2}, v_{2,2}, v_{1,2,2}, v_{2,2,2}, w_1, w_{1,2}, w_{2,2}, w_{1,2,2}, w_{2,2,2}\} \tag{16}$$

```
E >
E > {seq(coeffs(expand(deteqns[i]),split_vars),i=1..nops(deteqns))}
    :detsys:=simplify(%):nops(%);#indets(%%);
```

$$84 \tag{17}$$

```
E > # determining system of 84 equations
E >
E >
E > ### conditions on unknowns
E >
E > # at least one component of the symmetry must be non-zero
E >
E > symm_conds:=[add(symm_vars[i]^2,i=1..nops(symm_vars))<>0];
```

$$symm\_conds := \left[eta\_u^2 + eta\_v^2 + eta\_w^2 + \tau^2 + \xi^2 \neq 0\right] \tag{18}$$

```
E >
E >
E > # k(lambda) must be non-constant (otherwise the elastic string
    PDEs are linear)
E >
E > classify_vars:=[k(u[2],v[2],w[2])];
```

$$classify\_vars := [k] \tag{19}$$

```
E > classify_conds:=[diff(k(u[2],v[2],w[2]),u[2])^2 +diff(k(u[2],v
    [2],w[2]),v[2])^2+diff(k(u[2],v[2],w[2]),w[2])^2<>0];
```

$$classify\_conds := \left[{k_{u_2}}^2 + {k_{v_2}}^2 + {k_{w_2}}^2 \neq 0\right] \tag{20}$$

```
E >
E >
E >
E >
E > ##### STEP 3
E >
E > ### bring determining system into involutive form
E >
E > fullsys:=[detsys[],classify_conds[],symm_conds]:nops(%);
```

$$86 \tag{21}$$

```
E > fulldetsys:=sort([detsys[],k_eqns[],classify_conds[],symm_conds[]
    ],length):nops(%);
```

$$89 \tag{22}$$

```
E >
E > split_sys:=rifsimp(fulldetsys,[symm_vars[],classify_vars],
    casesplit):
E > split_sys[casecount];
```

$$7 \tag{23}$$

```
E >
E > seq(print(i,remove(has,split_sys[i][Solved],symm_vars)),i=1..
    split_sys[casecount]);
```

$$1, \left[k_{u_2} = \frac{k_{w_2}(1+u_2)}{w_2}, k_{v_2} = \frac{k_{w_2} v_2}{w_2}\right]$$

$$2, \left[k_{w_2,w_2,w_2} = \frac{2 k k_{w_2,w_2}{}^2 w_2^2 - k_{w_2,w_2} k_{w_2}{}^2 w_2^2 - k k_{w_2,w_2} k_{w_2} w_2 + k_{w_2}{}^3 w_2 - k k_{w_2}{}^2}{k_{w_2} k w_2^2}, k_{u_2} = \frac{k_{w_2}(1+u_2)}{w_2}, k_{v_2} = \frac{k_{w_2} v_2}{w_2}\right]$$

$$3, \left[k_{w_2,w_2} = \frac{1}{k u_2^2 w_2 + k v_2^2 w_2 + k w_2^3 + 2 w_2 k u_2 + k w_2}\left(k_{w_2}{}^2 w_2^3 + k_{w_2}{}^2 w_2 + k k_{w_2} + k_{w_2}{}^2 u_2^2 w_2 + k k_{w_2} u_2^2 - k w_2^2 k_{w_2} + 2 k_{w_2}{}^2 u_2 w_2 + k_{w_2}{}^2 v_2^2 w_2 + k_{w_2} k v_2^2 + 2 k k_{w_2} u_2\right), k_{u_2} = \frac{k_{w_2}(1+u_2)}{w_2}, k_{v_2} = \frac{k_{w_2} v_2}{w_2}\right]$$

$$4, \left[k_{w_2,w_2} = \frac{k_{w_2}\left(3 k_{w_2} w_2 + 2 k\right)}{2 k w_2}, k_{u_2} = \frac{k_{w_2}(1+u_2)}{w_2}, k_{v_2} = \frac{k_{w_2} v_2}{w_2}\right]$$

$$5, \left[k_{u_2} = -\frac{4(1+u_2) k}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}, k_{v_2} = -\frac{4 k v_2}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}, k_{w_2} = \frac{4 k w_2}{-u_2^2 - v_2^2 - w_2^2 - 2 u_2 - 1}\right]$$

$$6, \left[k_{w_2,w_2} = \frac{k_{w_2}\left(3 k_{w_2} w_2 + k\right)}{k w_2}, k_{u_2} = \frac{k_{w_2}(1+u_2)}{w_2}, k_{v_2} = \frac{k_{w_2} v_2}{w_2}\right]$$

$$7, \left[k_{u_2} = -\frac{(1+u_2) k}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}, k_{v_2} = -\frac{k v_2}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}, k_{w_2} = -\frac{k w_2}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}\right] \tag{24}$$

```
E >
E >
```

```
E > ### remove redundant cases
E > dsubs(split_sys[2][Solved],split_sys[1][Solved]):{%[]} minus
    {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);
```

$$\varnothing \tag{25}$$

```
E > # 2 is special case of 1
E >
E > dsubs(split_sys[3][Solved],split_sys[1][Solved]):{%[]} minus
    {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);
```

$$\left\{t, x, u[\,], u_2, v[\,], v_2, w[\,], w_2, k_{w_2}, \tau_t, \xi_x, k, \tau, \xi\right\} \tag{26}$$

```
E > # 3 is new
E >
E > dsubs(split_sys[4][Solved],split_sys[1][Solved]):{%[]} minus
    {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);
```

$$\varnothing \tag{27}$$

```
E >
E >
E > # 4 is special case of 1
E >
E > dsubs(split_sys[5][Solved],split_sys[1][Solved]):{%[]} minus
    {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))};indets(%);
```

$$\left\{0, \tau_t - \xi_x, \frac{\tau_t}{2} - \frac{\xi_x}{2}\right\}$$

$$\left\{t, x, u[\,], v[\,], w[\,], \tau_t, \xi_x, \tau, \xi\right\} \tag{28}$$

```
E > dsubs(split_sys[5][Solved],split_sys[3][Solved]):{%[]} minus
    {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);
```

$$\left\{t, x, u[\,], v[\,], w[\,], \tau_{t,t}, \tau_t, \tau\right\} \tag{29}$$

```
E > # 5 is new
E >
E > dsubs(split_sys[6][Solved],split_sys[1][Solved]):{%[]} minus
    {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))};indets(%);
```

$$\{0\}$$

$$\varnothing \tag{30}$$

```
E > # 6 is special case of 1
E >
E > dsubs(split_sys[7][Solved],split_sys[1][Solved]):{%[]} minus
    {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))};indets(%);
```

$$\left\{0, \tau_t - \xi_x, 2\,\tau_t - 2\,\xi_x\right\}$$

$$\left\{t, x, u[\,], v[\,], w[\,], \tau_t, \xi_x, \tau, \xi\right\} \tag{31}$$

```
E > dsubs(split_sys[7][Solved],split_sys[3][Solved]):{%[]} minus
    {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))};indets(%);
```

$$\{0\}$$

$$\varnothing \tag{32}$$

```
E > # 7 is special case of 3
E >
E >
E >
E > ### List of solution CASES: 1,3,5
E >
E >
E > ##### STEP 4
E >
E > ### Do for each case:

E > # first look at the case output and see if it contains a
    constraint
E > # next look at case equations and separate into algebraic
    conditions, ODEs, PDEs
E > # solve for k first
E > # solve simplest eqns (...=0) next
E > # solve remaining eqns
E >
E >
E >
E > ### CASE 1
E >
E > split_sys[1][Constraint];
```

$$\left(split\_sys_1\right)_{Constraint} \tag{33}$$

```
E > split_sys[1][Solved]:sys1:=simplify(%);nops(%);
```

$$sys1 := \left[eta\_u_{t,t} = 0, eta\_u_x = 0, eta\_u_{u[\,]} = \xi_x, eta\_u_{v[\,]} = -eta\_v_{u[\,]}, eta\_u_{w[\,]} = -eta\_w_{u[\,]}, eta\_v_{t,t} = 0, eta\_v_{t,u[\,]} = 0, eta\_v_{u[\,],u[\,]} = 0, eta\_v_x = eta\_v_{u[\,]}, eta\_v_{v[\,]} = \xi_x, eta\_v_{w[\,]} = -eta\_w_{v[\,]}, eta\_w_{t,t} = 0, eta\_w_{t,u[\,]} = 0, eta\_w_{t,v[\,]} = 0, eta\_w_{u[\,],u[\,]} = 0, eta\_w_{u[\,],v[\,]} = 0, eta\_w_{v[\,],v[\,]} = 0, eta\_w_x = eta\_w_{u[\,]}, eta\_w_{w[\,]} = \xi_x, \tau_t = \xi_x, \tau_x = 0, \tau_{u[\,]} = 0, \tau_{v[\,]} = 0, \tau_{w[\,]} = 0, \xi_{x,x} = 0, \xi_t = 0, \xi_{u[\,]} = 0, \xi_{v[\,]} = 0,\right.$$

$$\xi_{w[\,]} = 0,\; k_{u_2} = \frac{k_{w_2}\left(1+u_2\right)}{w_2},\; k_{v_2} = \frac{k_{w_2}\, v_2}{w_2}\Bigg]$$

$$31 \tag{34}$$

E >

E > `remove(has,sys1,symm_vars);dsubs(k_funct,%):{seq(simplify(lhs(%[i])-rhs(%[i])),i=1..nops(%))};`

$$\left[k_{u_2} = \frac{k_{w_2}\left(1+u_2\right)}{w_2},\; k_{v_2} = \frac{k_{w_2}\, v_2}{w_2}\right]$$

$$\{0\} \tag{35}$$

E > `k_soln1:=k_funct;`

$$k\_soln1 := k = k\left(\sqrt{\left(1+u_2\right)^2 + v_2^2 + w_2^2} - 1\right) \tag{36}$$

E > `remove(has,sys1,k(u[2],v[2],w[2])):sys1eqns1:=sort(%,length);nops(%);`

$$sys1eqns1 := \left[\xi_t = 0,\, \tau_x = 0,\, eta\_u_x = 0,\, \xi_{u[\,]} = 0,\, \xi_{v[\,]} = 0,\, \xi_{w[\,]} = 0,\, \tau_{u[\,]} = 0,\, \tau_{v[\,]} = 0,\, \tau_{w[\,]} = 0,\, \xi_{x,x} = 0,\right.$$
$$eta\_u_{t,t} = 0,\, eta\_v_{t,t} = 0,\, eta\_w_{t,t} = 0,\, eta\_v_{t,u[\,]} = 0,\, eta\_w_{t,u[\,]} = 0,\, eta\_w_{t,v[\,]} = 0,\, eta\_v_{u[\,],u[\,]} = 0,$$
$$eta\_w_{u[\,],u[\,]} = 0,\, eta\_w_{u[\,],v[\,]} = 0,\, eta\_w_{v[\,],v[\,]} = 0,\, \tau_t = \xi_x,\, eta\_u_{u[\,]} = \xi_x,\, eta\_v_{v[\,]} = \xi_x,\, eta\_w_{w[\,]}$$
$$= \xi_x,\, eta\_v_x = eta\_v_{u[\,]},\, eta\_w_x = eta\_w_{u[\,]},\, eta\_u_{v[\,]} = -eta\_v_{u[\,]},\, eta\_u_{w[\,]} = -eta\_w_{u[\,]},\, eta\_v_{w[\,]} =$$
$$\left. -eta\_w_{v[\,]}\right]$$

$$29 \tag{37}$$

E >

E > `select(has,sys1eqns1,0):pdsolve(%):sys1soln1:=simplify(%);`

$$sys1soln1 := \left\{eta\_u = f_{15}(u[\,], v[\,], w[\,])\, t + f_{16}(u[\,], v[\,], w[\,]),\, eta\_v = f_{12}(x, v[\,], w[\,])\, t + f_{13}(x,\right.$$
$$v[\,], w[\,])\, u[\,] + f_{14}(x, v[\,], w[\,]),\, eta\_w = f_3(x, w[\,])\, t + f_6(x, w[\,])\, u[\,] + f_7(x, w[\,])\, v[\,] + f_8(x,$$
$$\left. w[\,]),\, \tau = f_9(t),\, \xi = c_1 x + c_2\right\} \tag{38}$$

E > `map(rhs,sys1soln1):indets(%,name) minus indets(symm_vars,name):sys1Cs1:=%;indets(%%%,name) minus indets(indets(%%%,function),name) minus %;`

$$sys1Cs1 := \{c_1, c_2\}$$

$$\varnothing \tag{39}$$

E > `dsubs(sys1soln1,sys1eqns1):{seq(coeffs(expand(lhs(%[i])-rhs(%[i])),[]),i=1..nops(%))}:rifsimp(%,casesplit):simplify(%[Solved]):sys1eqns2:=%;nops(%);`

$$sys1eqns2 := \left[f_{12_x} = 0,\, f_{13_x} = 0,\, f_{14_x} = f_{13}(x, v[\,], w[\,]),\, f_{3_x} = 0,\, f_{6_x} = 0,\, f_{7_x} = 0,\, f_{8_x} = f_6(x, w[\,]),\, f_{15_{u[\,]}} = 0,\right.$$

$$f_{16_{u[\,]}} = c_1, f_{12_{v[\,]}} = 0, f_{13_{v[\,]}} = 0, f_{14_{v[\,]}} = c_1, f_{15_{v[\,]}} = 0, f_{16_{v[\,]}} = -f_{13}(x, v[\,], w[\,]), f_{12_{w[\,]}} = 0, f_{13_{w[\,]}} = 0, f_{14_{w[\,]}} = -f_7(x, w[\,]), f_{15_{w[\,]}} = 0, f_{16_{w[\,]}} = -f_6(x, w[\,]), f_{3_{w[\,]}} = 0, f_{6_{w[\,]}} = 0, f_{7_{w[\,]}} = 0, f_{8_{w[\,]}} = c_1, f_{9_t} = c_1\Big]$$

$$24 \quad \textbf{(40)}$$

```
E > select(has,sys1eqns2,0);pdsolve(%):subs({_C1=_C11,_C2=_C12},%)
    :sys1soln2:=simplify(%);
```

$$\left[f_{12_x} = 0, f_{13_x} = 0, f_{3_x} = 0, f_{6_x} = 0, f_{7_x} = 0, f_{15_{u[\,]}} = 0, f_{12_{v[\,]}} = 0, f_{13_{v[\,]}} = 0, f_{15_{v[\,]}} = 0, f_{12_{w[\,]}} = 0, f_{13_{w[\,]}} = 0, f_{15_{w[\,]}} = 0, f_{3_{w[\,]}} = 0, f_{6_{w[\,]}} = 0, f_{7_{w[\,]}} = 0\right]$$

$$sys1soln2 := \{f_{12}(x, v[\,], w[\,]) = \_C11, f_{13}(x, v[\,], w[\,]) = \_C12, f_{15}(u[\,], v[\,], w[\,]) = c_3, f_3(x, w[\,]) = c_4, f_6(x, w[\,]) = c_5, f_7(x, w[\,]) = c_6\} \quad \textbf{(41)}$$

```
E > map(rhs,sys1soln2):indets(%,name) minus indets(symm_vars,name)
    :sys1Cs2:=% union sys1Cs1;indets(%%%,name) minus indets(indets
    (%%%,function),name) minus %;
```

$$sys1Cs2 := \{\_C11, \_C12, c_1, c_2, c_3, c_4, c_5, c_6\}$$

$$\varnothing \quad \textbf{(42)}$$

```
E > dsubs(sys1soln2,sys1eqns2):{seq(coeffs(expand(lhs(%[i])-rhs(%[i])
    ),[]),i=1..nops(%))}:rifsimp(%,casesplit):simplify(%[Solved])
    :sys1eqns3:=%;nops(%);
```

$$sys1eqns3 := \left[f_{14_x} = \_C12, f_{8_x} = c_5, f_{16_{u[\,]}} = c_1, f_{14_{v[\,]}} = c_1, f_{16_{v[\,]}} = -\_C12, f_{14_{w[\,]}} = -c_6, f_{16_{w[\,]}} = -c_5, f_{8_{w[\,]}} = c_1, f_{9_t} = c_1\right]$$

$$9 \quad \textbf{(43)}$$

```
E > sys1eqns3:pdsolve(%):sys1soln3:=simplify(%);
```

$$sys1soln3 := \{f_{14}(x, v[\,], w[\,]) = c_1 v[\,] + c_{12} x - c_6 w[\,] + c_{13}, f_{16}(u[\,], v[\,], w[\,]) = c_1 u[\,] - c_{12} v[\,] - c_5 w[\,] + c_{14}, f_8(x, w[\,]) = c_1 w[\,] + c_5 x + c_{15}, f_9(t) = c_1 t + c_{16}\} \quad \textbf{(44)}$$

```
E > map(rhs,sys1soln3):indets(%,name) minus indets(symm_vars,name)
    :sys1Cs3:=% union sys1Cs2;#indets(%%%,name) minus indets(indets
    (%%%,function),name) minus %;
```

$$sys1Cs3 := \{\_C11, c_1, c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_2, c_3, c_4, c_5, c_6\} \quad \textbf{(45)}$$

```
E > # all eqns solved
E > sys1soln1:dsubs(sys1soln2,%):dsubs(sys1soln3,%):soln1:=simplify(
    [%[]]);
```

$$soln1 := \left[eta\_u = c_1 u[\,] - c_{12} v[\,] + c_3 t - c_5 w[\,] + c_{14}, eta\_v = \left(x + u[\,]\right) c_{12} + \_C11\, t - c_6 w[\,] + c_1 v[\,] + c_{13}, eta\_w = \left(x + u[\,]\right) c_5 + c_4 t + c_1 w[\,] + c_6 v[\,] + c_{15}, \tau = c_1 t + c_{16}, \xi = c_1 x + c_2\right] \quad \textbf{(46)}$$

```
E >
E > dsubs(soln1,detsys):dsubs(k_soln1,%);
```

$$\{0\} \quad (47)$$

```
E > # verified
E >
E > map(rhs,soln1):indets(%,name) minus indets(symm_vars,name)
    :soln1Cs:=%;nops(%);
```

$$soln1Cs := \{\_C11, c_1, c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_2, c_3, c_4, c_5, c_6\}$$

$$12 \quad (48)$$

```
E >
E > k_soln1:subs(solve(strain,{w[2]^2}),rhs(%)):simplify(%) assuming
    (lambda>0);
```

Warning, solving for expressions other than names or functions is not recommended.

$$k(\lambda) \quad (49)$$

```
E >
E >
E >
E > ### CASE 3
E > split_sys[3][Pivots]:sys3conds:=%;
```

$$sys3conds := \left[k_{w_2} \neq 0, eta\_u^2 + eta\_v^2 + eta\_w^2 + \tau^2 + \xi^2 \neq 0, k_{w_2} u_2^2 + k_{w_2} v_2^2 + k_{w_2} w_2^2 + k w_2 + 2 k_{w_2} u_2 + k_{w_2} \neq 0, k_{w_2} u_2^2 + k_{w_2} v_2^2 + k_{w_2} w_2^2 + 2 k_{w_2} u_2 + 4 k w_2 + k_{w_2} \neq 0\right] \quad (50)$$

```
E > split_sys[3][Constraint];
```

$$(split\_sys_3)_{Constraint} \quad (51)$$

```
E > split_sys[3][Solved]:sys3:=simplify(%);nops(%);
```

$$sys3 := \Bigg[eta\_u_{t,t} = 0, eta\_u_x = \frac{2\left(\xi_x - \tau_t\right) k w_2}{k_{w_2}\left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1\right)}, eta\_u_{u[\,]} = \frac{\left(k_{w_2}\left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1\right) + 2 k w_2\right)\xi_x - 2k\tau_t w_2}{k_{w_2}\left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1\right)}, eta\_u_{v[\,]} = -eta\_v_{u[\,]}, eta\_u_{w[\,]} = -eta\_w_{u[\,]}, eta\_v_{t,t} = 0, eta\_v_{t,u[\,]} = 0, eta\_v_{u[\,],u[\,]} = 0, eta\_v_x = eta\_v_{u[\,]}, eta\_v_{v[\,]} = \frac{\left(k_{w_2}\left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1\right) + 2 k w_2\right)\xi_x - 2k\tau_t w_2}{k_{w_2}\left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1\right)}, eta\_v_{w[\,]} = -eta\_w_{v[\,]}, eta\_w_{t,t} = 0,$$

$$eta\_w_{t,u[\,]} = 0,\ eta\_w_{t,v[\,]} = 0,\ eta\_w_{u[\,],u[\,]} = 0,\ eta\_w_{u[\,],v[\,]} = 0,\ eta\_w_{v[\,],v[\,]} = 0,\ eta\_w_x = eta\_w_{u[\,]},$$

$$eta\_w_{w[\,]} = \frac{\left(k_{w_2}\left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1\right) + 2k\,w_2\right)\xi_x - 2k\,\tau_t\,w_2}{k_{w_2}\left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1\right)},\ \tau_{t,t} = 0,\ \tau_x = 0,\ \tau_{u[\,]} = 0,\ \tau_{v[\,]}$$

$$= 0,\ \tau_{w[\,]} = 0,\ \xi_{x,x} = 0,\ \xi_t = 0,\ \xi_{u[\,]} = 0,\ \xi_{v[\,]} = 0,\ \xi_{w[\,]} = 0,\ k_{w_2,w_2}$$

$$= \frac{k_{w_2}\left(w_2\left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1\right)k_{w_2} + k\left(u_2^2 + v_2^2 - w_2^2 + 2u_2 + 1\right)\right)}{k\,w_2\left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1\right)},\ k_{u_2} = \frac{k_{w_2}\left(1 + u_2\right)}{w_2},$$

$$k_{v_2} = \frac{k_{w_2}\,v_2}{w_2}\Bigg]$$

32 **(52)**

```
E >
E > remove(has,sys3,symm_vars):dsubs(k_funct,%):{seq(simplify(lhs(%
    [i])-rhs(%[i])),i=1..nops(%))}:subs(solve(strain,{w[2]^2}),%)
    :simplify(%) assuming(lambda>0):convert(%,diff):{seq(coeffs
    (expand(%[i]),[u[2],v[2],w[2]]),i=1..nops(%))} minus {0}
    :sys3_keqns:=simplify(%);nops(%);
```

Warning, solving for expressions other than names or functions is not recommended.

$$sys3\_keqns := \left\{\frac{-2k(\lambda)\,(\lambda+1)\,k_{\lambda,\lambda} + 2k_\lambda\left((\lambda+1)\,k_\lambda - k(\lambda)\right)}{(\lambda+1)^3\,k(\lambda)},\right.$$

$$\frac{-k(\lambda)\,(\lambda+1)\,k_{\lambda,\lambda} + k_\lambda\left((\lambda+1)\,k_\lambda - k(\lambda)\right)}{(\lambda+1)^3\,k(\lambda)},$$

$$\left.\frac{\lambda\left(k(\lambda)\,(\lambda+1)\,k_{\lambda,\lambda} - k_\lambda\left((\lambda+1)\,k_\lambda - k(\lambda)\right)\right)(\lambda+2)}{(\lambda+1)^3\,k(\lambda)}\right\}$$

3 **(53)**

```
E > rifsimp({sys3_keqns[],diff(k(lambda),lambda)<>0},casesplit)
    :simplify(%[Solved]);dsolve(%)[]:subs({_C2=k0,_C1=p},%):sys3_k:=%
    [];
```

$$\left[k_{\lambda,\lambda} = \frac{k_{\lambda}\left(\left(\lambda+1\right)k_{\lambda} - k(\lambda)\right)}{k(\lambda)\left(\lambda+1\right)}\right]$$

$$sys3\_k := k(\lambda) = \left(\lambda+1\right)^{p} k0 \quad \textbf{(54)}$$

```
E > k_soln3:=k(u[2],v[2],w[2])=subs(strain,rhs(sys3_k));
```

$$k\_soln3 := k = \left(\sqrt{\left(1+u_2\right)^2 + v_2^2 + w_2^2}\right)^{p} k0 \quad \textbf{(55)}$$

```
E > dsubs(k_soln3,sys3conds):simplify(%):select(has,%,p):solve(%,{p})
    :sys3_p:=%;
```

$$sys3\_p := \{p \neq -4, p \neq -1, p \neq 0\} \quad \textbf{(56)}$$

```
E > select(has,sys3,symm_vars):dsubs(k_soln3,%):simplify(%)
    :sys3eqns1:=sort(%,length);nops(%);
```

$$sys3eqns1 := \Big[\xi_t = 0, \tau_x = 0, \xi_{u[\,]} = 0, \xi_{v[\,]} = 0, \xi_{w[\,]} = 0, \tau_{u[\,]} = 0, \tau_{v[\,]} = 0, \tau_{w[\,]} = 0, \xi_{x,x} = 0, \tau_{t,t} = 0,$$

$$eta\_u_{t,t} = 0, eta\_v_{t,t} = 0, eta\_w_{t,t} = 0, eta\_v_{t,u[\,]} = 0, eta\_w_{t,u[\,]} = 0, eta\_w_{t,v[\,]} = 0, eta\_v_{u[\,],u[\,]} = 0,$$

$$eta\_w_{u[\,],u[\,]} = 0, eta\_w_{u[\,],v[\,]} = 0, eta\_w_{v[\,],v[\,]} = 0, eta\_v_x = eta\_v_{u[\,]}, eta\_w_x = eta\_w_{u[\,]}, eta\_u_{v[\,]} =$$

$$-eta\_v_{u[\,]}, eta\_u_{w[\,]} = -eta\_w_{u[\,]}, eta\_v_{w[\,]} = -eta\_w_{v[\,]}, eta\_u_x = \frac{2\,\xi_x - 2\,\tau_t}{p}, eta\_u_{u[\,]}$$

$$= \frac{(p+2)\,\xi_x - 2\,\tau_t}{p}, eta\_v_{v[\,]} = \frac{(p+2)\,\xi_x - 2\,\tau_t}{p}, eta\_w_{w[\,]} = \frac{(p+2)\,\xi_x - 2\,\tau_t}{p}\Big]$$

$$29 \quad \textbf{(57)}$$

```
E > select(has,sys3eqns1,0):pdsolve({%[],sys3_p[]}):sys3soln1:=
    simplify(%);
```

$$sys3soln1 := \Big\{eta\_u = f_{14}(x, u[\,], v[\,], w[\,])\,t + f_{15}(x, u[\,], v[\,], w[\,]), eta\_v = f_{11}(x, v[\,], w[\,])\,t \quad \textbf{(58)}$$

$$+ f_{12}(x, v[\,], w[\,])\,u[\,] + f_{13}(x, v[\,], w[\,]), eta\_w = f_3(x, w[\,])\,t + f_6(x, w[\,])\,u[\,] + f_7(x, w[\,])\,v[\,]$$

$$+ f_8(x, w[\,]), \tau = c_1\,t + c_2, \xi = c_3\,x + c_4\Big\}$$

```
E > map(rhs,sys3soln1):indets(%,name) minus indets(symm_vars,name)
    :sys3Cs1:=%;indets(%%%,name) minus indets(indets(%%%,function),
    name) minus %;
```

$$sys3Cs1 := \{c_1, c_2, c_3, c_4\}$$

$$\{t\} \quad \textbf{(59)}$$

```
E > dsubs(sys3soln1,sys3eqns1):{seq(coeffs(expand(lhs(%[i])-rhs(%[i])
    ),[t]),i=1..nops(%))}:rifsimp({%[],sys3_p[]},casesplit):simplify
    (%[Solved]):sys3eqns2:=%;nops(%);
```

$$sys3eqns2 := \Big[f_{11_x} = 0, f_{12_x} = 0, f_{13_x} = f_{12}(x, v[\,], w[\,]), f_{14_x} = 0, f_{15_x} = \frac{2\,c_3 - 2\,c_1}{p}, f_{3_x} = 0, f_{6_x} = 0, f_{7_x} = 0,$$

$$f_{8_x} = f_6(x, w[\,]), f_{14_{u[\,]}} = 0, f_{15_{u[\,]}} = \frac{(p+2)\,c_3 - 2\,c_1}{p}, f_{11_{v[\,]}} = 0, f_{12_{v[\,]}} = 0, f_{13_{v[\,]}} = \frac{(p+2)\,c_3 - 2\,c_1}{p}, f_{14_{v[\,]}} = 0, f_{15_{v[\,]}} = -f_{12}(x, v[\,], w[\,]), f_{11_{w[\,]}} = 0, f_{12_{w[\,]}} = 0, f_{13_{w[\,]}} = -f_7(x, w[\,]), f_{14_{w[\,]}} = 0, f_{15_{w[\,]}} = -f_6(x, w[\,]), f_{3_{w[\,]}} = 0, f_{6_{w[\,]}} = 0, f_{7_{w[\,]}} = 0, f_{8_{w[\,]}} = \frac{(p+2)\,c_3 - 2\,c_1}{p}\Big]$$

25 **(60)**

```
E > select(has,sys3eqns2,0);pdsolve(%):subs({_C1=_C11,_C2=_C12,_C3=
    _C13,_C4=_C14},%):sys3soln2:=simplify(%);
```

$$\left[f_{11_x} = 0, f_{12_x} = 0, f_{14_x} = 0, f_{3_x} = 0, f_{6_x} = 0, f_{7_x} = 0, f_{14_{u[\,]}} = 0, f_{11_{v[\,]}} = 0, f_{12_{v[\,]}} = 0, f_{14_{v[\,]}} = 0, f_{11_{w[\,]}} = 0, f_{12_{w[\,]}} = 0, f_{14_{w[\,]}} = 0, f_{3_{w[\,]}} = 0, f_{6_{w[\,]}} = 0, f_{7_{w[\,]}} = 0\right]$$

$$sys3soln2 := \{f_{11}(x, v[\,], w[\,]) = c_{12}, f_{12}(x, v[\,], w[\,]) = c_{13}, f_{14}(x, u[\,], v[\,], w[\,]) = \_C11, f_3(x, w[\,]) = c_{14}, f_6(x, w[\,]) = c_5, f_7(x, w[\,]) = c_6\} \tag{61}$$

```
E > map(rhs,sys3soln2):indets(%,name) minus indets(symm_vars,name)
    :sys3Cs2:=% union sys3Cs1;indets(%%%,name) minus indets(indets
    (%%%,function),name) minus %;
```

$$sys3Cs2 := \{\_C11, c_1, c_{12}, c_{13}, c_{14}, c_2, c_3, c_4, c_5, c_6\}$$

$$\varnothing \tag{62}$$

```
E > dsubs(sys3soln2,sys3eqns2):{seq(coeffs(expand(lhs(%[i])-rhs(%[i])
    ),[t]),i=1..nops(%))}:rifsimp({%[],sys3_p[]},casesplit):simplify
    (%[Solved]):sys3eqns3:=%;nops(%);
```

$$sys3eqns3 := \left[f_{13_x} = c_{13}, f_{15_x} = \frac{2\,c_3 - 2\,c_1}{p}, f_{8_x} = c_5, f_{15_{u[\,]}} = \frac{(p+2)\,c_3 - 2\,c_1}{p}, f_{13_{v[\,]}} = \frac{(p+2)\,c_3 - 2\,c_1}{p}, f_{15_{v[\,]}} = -c_{13}, f_{13_{w[\,]}} = -c_6, f_{15_{w[\,]}} = -c_5, f_{8_{w[\,]}} = \frac{(p+2)\,c_3 - 2\,c_1}{p}\right]$$

$$9 \tag{63}$$

```
E > sys3eqns3:pdsolve({%[],sys3_p[]}):subs({_C14=_C24},%):sys3soln3:=
    simplify(%);
```

$$sys3soln3 := \left\{f_{13}(x, v[\,], w[\,]) = c_{13}\,x - c_6\,w[\,] + \frac{\left((p+2)\,c_3 - 2\,c_1\right)v[\,]}{p} + c_{15}, f_{15}(x, u[\,], v[\,], w[\,]) = \frac{\left(-c_{13}\,v[\,] + c_3\,u[\,] - c_5\,w[\,] + \_C24\right)p - 2\left(-c_3 + c_1\right)\left(x + u[\,]\right)}{p}, f_8(x, w[\,]) = c_5\,x + \frac{\left((p+2)\,c_3 - 2\,c_1\right)w[\,]}{p} + c_{16}\right\} \tag{64}$$

```
E > map(rhs,sys3soln3):indets(%,name) minus indets({k_soln3,
    symm_vars},name):sys3Cs3:=% union sys3Cs2;
```

$$sys3Cs3 := \{\_C11, \_C24, c_1, c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_2, c_3, c_4, c_5, c_6\} \quad \mathbf{(65)}$$

```
E > # all eqns solved
E >
E > sys3soln1:dsubs(sys3soln2,%):dsubs(sys3soln3,%):soln3:=simplify(
    [%[]]);
```

$$soln3 := \left[eta\_u = \frac{\left(\_C11\, t - c_{13} v[\,] + c_3 u[\,] - c_5 w[\,] + \_C24\right) p - 2\left(-c_3 + c_1\right)\left(x + u[\,]\right)}{p}, eta\_v = \frac{\left(c_3 v[\,] + \left(x + u[\,]\right) c_{13} - c_6 w[\,] + c_{12} t + c_{15}\right) p - 2\, v[\,]\left(-c_3 + c_1\right)}{p}, eta\_w = \frac{\left(c_3 w[\,] + \left(x + u[\,]\right) c_5 + c_6 v[\,] + c_{14} t + c_{16}\right) p - 2\, w[\,]\left(-c_3 + c_1\right)}{p}, \tau = c_1 t + c_2, \xi = c_3 x + c_4\right] \quad \mathbf{(66)}$$

```
E > dsubs(soln3,detsys):dsubs(k_soln3,%);
```

$$\{0\} \quad \mathbf{(67)}$$

```
E > # verified
E >
E > map(rhs,soln3):indets(%,name) minus indets({symm_vars,k_soln3},
    name):soln3Cs:=%;nops(%);
```

$$soln3Cs := \{\_C11, \_C24, c_1, c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_2, c_3, c_4, c_5, c_6\}$$

$$13 \quad \mathbf{(68)}$$

```
E >
E > k_soln3;sys3_p;
```

$$k = \left(\sqrt{(1 + u_2)^2 + v_2^2 + w_2^2}\right)^p k0$$

$$\{p \neq -4, p \neq -1, p \neq 0\} \quad \mathbf{(69)}$$

```
E >
E >
E >
E > ### CASE 5
E >
E > split_sys[5][Pivots]:sys5conds:=%;
```

$$sys5conds := \left[k \neq 0, eta\_u^2 + eta\_v^2 + eta\_w^2 + \tau^2 + \xi^2 \neq 0\right] \quad \mathbf{(70)}$$

```
E > split_sys[5][Constraint];
```

$$\left(split\_sys_5\right)_{Constraint} \quad \mathbf{(71)}$$

```
E > split_sys[5][Solved]:sys5:=simplify(%);nops(%);
```

$$sys5 := \left[eta\_u_{t,t} = 0, eta\_u_x = \frac{\tau_t}{2} - \frac{\xi_x}{2}, eta\_u_{u[\,]} = \frac{\tau_t}{2} + \frac{\xi_x}{2}, eta\_u_{v[\,]} = -eta\_v_{u[\,]}, eta\_u_{w[\,]} =\right.$$

$$-eta\_w_{u[\,]}, eta\_v_{t,t} = 0, eta\_v_{t,u[\,]} = 0, eta\_v_{u[\,],u[\,]} = 0, eta\_v_x = eta\_v_{u[\,]}, eta\_v_{v[\,]} = \frac{\tau_t}{2} + \frac{\xi_x}{2},$$

$$eta\_v_{w[\,]} = -eta\_w_{v[\,]}, eta\_w_{t,t} = 0, eta\_w_{t,u[\,]} = 0, eta\_w_{t,v[\,]} = 0, eta\_w_{u[\,],u[\,]} = 0, eta\_w_{u[\,],v[\,]} = 0,$$

$$eta\_w_{v[\,],v[\,]} = 0, eta\_w_x = eta\_w_{u[\,]}, eta\_w_{w[\,]} = \frac{\tau_t}{2} + \frac{\xi_x}{2}, \tau_{t,t,t} = 0, \tau_x = 0, \tau_{u[\,]} = 0, \tau_{v[\,]} = 0, \tau_{w[\,]}$$

$$= 0, \xi_{x,x} = 0, \xi_t = 0, \xi_{u[\,]} = 0, \xi_{v[\,]} = 0, \xi_{w[\,]} = 0, k_{u_2} = -\frac{4\left(1 + u_2\right) k}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}, k_{v_2} =$$

$$\left. -\frac{4 k v_2}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}, k_{w_2} = -\frac{4 k w_2}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1} \right]$$

$$32 \tag{72}$$

```
E > remove(has,sys5,symm_vars):dsubs(k_funct,%):{seq(simplify(lhs(%
    [i])-rhs(%[i])),i=1..nops(%))}:subs(solve(strain,{w[2]^2}),%)
    :simplify(%) assuming(lambda>0):convert(%,diff):{seq(coeffs
    (expand(%[i]),[u[2],v[2],w[2]]),i=1..nops(%))} minus {0}
    :sys5_keqns:=simplify(%);nops(%);
```

Warning, solving for expressions other than names or functions is not recommended.

$$sys5\_keqns := \left\{ \frac{(\lambda + 1)\, k_\lambda + 4\, k(\lambda)}{(\lambda + 1)^2} \right\}$$

$$1 \tag{73}$$

```
E > rifsimp({sys5_keqns[],diff(k(lambda),lambda)<>0},casesplit)
    :simplify(%[Solved]);dsolve(%)[]:subs({_C1=k0},%):sys5_k:=%;
```

$$\left[k_\lambda = -\frac{4\, k(\lambda)}{\lambda + 1} \right]$$

$$sys5\_k := k(\lambda) = \frac{k0}{(\lambda + 1)^4} \tag{74}$$

```
E > k_soln5:=k(u[2],v[2],w[2])=subs(strain,rhs(sys5_k));
```

$$k\_soln5 := k = \frac{k0}{\left((1 + u_2)^2 + v_2^2 + w_2^2\right)^2} \tag{75}$$

```
E > dsubs(k_soln5,sys5conds):simplify(%):select(has,%,k0);
```

$$\left[\frac{k0}{\left(u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1\right)^2} \neq 0 \right] \tag{76}$$

```
E >
```

```
E > select(has,sys5,symm_vars):simplify(%):sys5eqns1:=sort(%,length);
    nops(%);
```

$$sys5eqns1 := \left[\xi_t = 0, \tau_x = 0, \xi_{u[\,]} = 0, \xi_{v[\,]} = 0, \xi_{w[\,]} = 0, \tau_{u[\,]} = 0, \tau_{v[\,]} = 0, \tau_{w[\,]} = 0, \xi_{x,x} = 0, eta\_u_{t,t} = 0, eta\_v_{t,t} = 0, eta\_w_{t,t} = 0, eta\_v_{t,u[\,]} = 0, eta\_w_{t,u[\,]} = 0, eta\_w_{t,v[\,]} = 0, eta\_v_{u[\,],u[\,]} = 0, eta\_w_{u[\,],u[\,]} = 0, eta\_w_{u[\,],v[\,]} = 0, eta\_w_{v[\,],v[\,]} = 0, \tau_{t,t,t} = 0, eta\_v_x = eta\_v_{u[\,]}, eta\_w_x = eta\_w_{u[\,]}, eta\_u_{v[\,]} = -eta\_v_{u[\,]}, eta\_u_{w[\,]} = -eta\_w_{u[\,]}, eta\_v_{w[\,]} = -eta\_w_{v[\,]}, eta\_u_x = \frac{\tau_t}{2} - \frac{\xi_x}{2}, eta\_u_{u[\,]} = \frac{\tau_t}{2} + \frac{\xi_x}{2}, eta\_v_{v[\,]} = \frac{\tau_t}{2} + \frac{\xi_x}{2}, eta\_w_{w[\,]} = \frac{\tau_t}{2} + \frac{\xi_x}{2}\right]$$

$$29 \tag{77}$$

```
E > select(has,sys5eqns1,0):pdsolve(%):sys5soln1:=simplify(%);
```

$$sys5soln1 := \left\{eta\_u = f_{14}(x, u[\,], v[\,], w[\,])\, t + f_{15}(x, u[\,], v[\,], w[\,]), eta\_v = f_{11}(x, v[\,], w[\,])\, t + f_{12}(x, v[\,], w[\,])\, u[\,] + f_{13}(x, v[\,], w[\,]), eta\_w = f_3(x, w[\,])\, t + f_6(x, w[\,])\, u[\,] + f_7(x, w[\,])\, v[\,] + f_8(x, w[\,]), \tau = \frac{1}{2} c_1 t^2 + c_2 t + c_3, \xi = c_4 x + c_5\right\} \tag{78}$$

```
E > map(rhs,sys5soln1):indets(%,name) minus indets(symm_vars,name)
    :sys5Cs:=%;indets(%%%,name) minus indets(indets(%%%,function),
    name) minus %;
```

$$sys5Cs := \{c_1, c_2, c_3, c_4, c_5\}$$

$$\{t\} \tag{79}$$

```
E >
E > dsubs(sys5soln1,sys5eqns1):{seq(coeffs(expand(lhs(%[i])-rhs(%[i])
    ),[t]),i=1..nops(%))}:rifsimp({%[]},casesplit):simplify(%[Solved]
    ):sys5eqns2:=%;nops(%);
```

$$sys5eqns2 := \left[f_{11_x} = 0, f_{12_x} = 0, f_{13_x} = f_{12}(x, v[\,], w[\,]), f_{14_x} = \frac{c_1}{2}, f_{15_x} = \frac{c_2}{2} - \frac{c_4}{2}, f_{3_x} = 0, f_{6_x} = 0, f_{7_x} = 0, f_{8_x} = f_6(x, w[\,]), f_{14_{u[\,]}} = \frac{c_1}{2}, f_{15_{u[\,]}} = \frac{c_2}{2} + \frac{c_4}{2}, f_{11_{v[\,]}} = \frac{c_1}{2}, f_{12_{v[\,]}} = 0, f_{13_{v[\,]}} = \frac{c_2}{2} + \frac{c_4}{2}, f_{14_{v[\,]}} = 0, f_{15_{v[\,]}} = -f_{12}(x, v[\,], w[\,]), f_{11_{w[\,]}} = 0, f_{12_{w[\,]}} = 0, f_{13_{w[\,]}} = -f_7(x, w[\,]), f_{14_{w[\,]}} = 0, f_{15_{w[\,]}} = \right.$$

$$-f_6(x, w[\,]), f_{3_{w[\,]}} = \frac{c_1}{2}, f_{6_{w[\,]}} = 0, f_{7_{w[\,]}} = 0, f_{8_{w[\,]}} = \frac{c_2}{2} + \frac{c_4}{2}\Big]$$

$$25 \tag{80}$$

```
E > select(has,sys5eqns2,0);pdsolve(%):subs({_C1=_C11,_C2=_C12,_C3=
    _C13,_F15(v[])=_F25(v[])},%):sys5soln2:=%;
```

$$\left[f_{11_x} = 0, f_{12_x} = 0, f_{3_x} = 0, f_{6_x} = 0, f_{7_x} = 0, f_{12_{v[\,]}} = 0, f_{14_{v[\,]}} = 0, f_{11_{w[\,]}} = 0, f_{12_{w[\,]}} = 0, f_{14_{w[\,]}} = 0, f_{6_{w[\,]}} = 0, f_{7_{w[\,]}} = 0\right]$$

$$sys5soln2 := \{f_{11}(x, v[\,], w[\,]) = \_F25(v[\,]), f_{12}(x, v[\,], w[\,]) = \_C11, f_{14}(x, u[\,], v[\,], w[\,]) = f_{16}(x, u[\,]), f_3(x, w[\,]) = f_{17}(w[\,]), f_6(x, w[\,]) = c_{12}, f_7(x, w[\,]) = c_{13}\} \tag{81}$$

```
E > map(rhs,sys5soln2):indets(%,name) minus indets(symm_vars,name)
    :sys5Cs1:=%;indets(%%%,name) minus indets(indets(%%%,function),
    name) minus %;
```

$$sys5Cs1 := \{\_C11, c_{12}, c_{13}\}$$

$$\varnothing \tag{82}$$

```
E >
E > dsubs(sys5soln2,sys5eqns2):rifsimp({%[]},casesplit):simplify(%
    [Solved]):sys5eqns3:=%;nops(%);
```

$$sys5eqns3 := \left[f_{13_x} = \_C11, f_{15_x} = \frac{c_2}{2} - \frac{c_4}{2}, f_{16_x} = \frac{c_1}{2}, f_{8_x} = c_{12}, f_{15_{u[\,]}} = \frac{c_2}{2} + \frac{c_4}{2}, f_{16_{u[\,]}} = \frac{c_1}{2}, f_{13_{v[\,]}} = \frac{c_2}{2} + \frac{c_4}{2}, f_{15_{v[\,]}} = -\_C11, \_F25_{v[\,]} = \frac{c_1}{2}, f_{13_{w[\,]}} = -c_{13}, f_{15_{w[\,]}} = -c_{12}, f_{17_{w[\,]}} = \frac{c_1}{2}, f_{8_{w[\,]}} = \frac{c_2}{2} + \frac{c_4}{2}\right]$$

$$13 \tag{83}$$

```
E > pdsolve(sys5eqns3):sys5soln3:=simplify(%);
```

$$sys5soln3 := \left\{\_F25(v[\,]) = \frac{c_1 v[\,]}{2} + c_{21}, f_{13}(x, v[\,], w[\,]) = \frac{(c_2 + c_4)\, v[\,]}{2} + c_{18} x - c_{13} w[\,] + c_{15}, f_{15}(x, u[\,], v[\,], w[\,]) = \frac{(c_2 + c_4)\, u[\,]}{2} + \frac{x c_2}{2} - \frac{x c_4}{2} - c_{18} v[\,] - c_{12} w[\,] + c_{14}, f_{16}(x, u[\,]) = \frac{(x + u[\,])\, c_1}{2} + c_{19}, f_{17}(w[\,]) = \frac{c_1 w[\,]}{2} + c_{20}, f_8(x, w[\,]) = \frac{(c_2 + c_4)\, w[\,]}{2} + c_{12} x + c_{16}\right\} \tag{84}$$

```
E > map(rhs,sys5soln3):indets(%,name) minus indets({k_soln5,
    symm_vars},name):sys5Cs2:=% union sys5Cs1;
```

$$sys5Cs2 := \{c_1, c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_{18}, c_{19}, c_2, c_{20}, c_{21}, c_4\} \tag{85}$$

```
E > # all eqns solved
```

```
E >
E > sys5soln1:dsubs(sys5soln2,%):dsubs(sys5soln3,%):soln5:=simplify(
    [%[]]);
```

$$soln5 := \left[eta\_u = \frac{(c_1 t + c_2 + c_4)\, u[\,]}{2} + \frac{(c_1 x + 2 c_{19})\, t}{2} + \frac{(c_2 - c_4)\, x}{2} - c_{18} v[\,] - c_{12} w[\,] + c_{14}, eta\_v = \frac{(c_1 t + c_2 + c_4)\, v[\,]}{2} + (x + u[\,])\, c_{18} + t\, c_{21} - c_{13} w[\,] + c_{15}, eta\_w = \frac{(c_1 t + c_2 + c_4)\, w[\,]}{2} + (x + u[\,])\, c_{12} + t\, c_{20} + c_{13} v[\,] + c_{16}, \tau = \frac{1}{2} c_1 t^2 + c_2 t + c_3, \xi = x\, c_4 + c_5\right] \quad (86)$$

```
E > dsubs(soln5,detsys):dsubs(k_soln5,%);
```

$$\{0\} \quad (87)$$

```
E > # verified
E >
E > map(rhs,soln5):indets(%,name) minus indets({symm_vars,k_soln5},
    name):soln5Cs:=%;nops(%);
```

$$soln5Cs := \{c_1, c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_{18}, c_{19}, c_2, c_{20}, c_{21}, c_3, c_4, c_5\}$$

$$14 \quad (88)$$

```
E >
E > sys5_k
```

$$k(\lambda) = \frac{k0}{(\lambda + 1)^4} \quad (89)$$

```
E >
E >
E >
E > ######## Check variational symmetry condition
E >
E > L:=(rho*A/2)*(u[1]^2+v[1]^2+w[1]^2) - subs(strain,W(lambda));
```

$$L := \frac{\rho A \left(u_1^2 + v_1^2 + w_1^2\right)}{2} - W\left(\sqrt{(1 + u_2)^2 + v_2^2 + w_2^2} - 1\right) \quad (90)$$

```
E > EulerLagrange(L):collect(%,[u[1,1],v[1,1],w[1,1],u[2,2],v[2,2],w
    [2,2]],simplify):ELeqns:=%:
E > subs(lead_ders,ELeqns):dsubs(k_funct,%):simplify(%):algsubs(solve
    (strain,{w[2]^2})[],expand(%)):simplify(%) assuming(lambda>0)
    :convert(%,diff):{seq(coeffs(expand(%[i]),[u[2,2],v[2,2],w[2,2],u
    [2],v[2],w[2]]),i=1..nops(%))}:simplify(%):rifsimp(%,casesplit)
    :simplify(%[Solved]):W_eqn:=%[];
```

Warning, solving for expressions other than names or functions is not recommended.

$$W\_eqn := W_{\lambda} = A\,k(\lambda)\,\rho\,(\lambda+1) \tag{91}$$

```
E >
E >
E >
E > ### CASE 1
E > subs(soln1,P_u*eqn_u+P_v*eqn_v+P_w*eqn_w):subs(k_funct,%)
    :char_eqn1:=simplify(%):
E > EulerLagrange(char_eqn1):simplify(%):algsubs(solve(strain,{w[2]
    ^2})[],expand(%)):simplify(%) assuming(lambda>0):convert(%,diff)
    :remove(has,indets(%,name),{lambda,soln1Cs[]});{seq(coeffs(expand
    (%%[i]),%),i=1..nops(%%))}:simplify(%):case1_varconds:=%;
```

Warning, solving for expressions other than names or functions is not recommended.

$$\{u_2, u_{1,1}, u_{2,2}, v_2, v_{1,1}, v_{2,2}, w_2, w_{1,1}, w_{2,2}\}$$

$$case1\_varconds := \left\{2\,c_1, -2\,c_1\,k(\lambda), -\frac{4\,c_1\,k_{\lambda}}{\lambda+1}, -\frac{2\,c_1\,k_{\lambda}}{\lambda+1}, \frac{2\,c_1\,k_{\lambda}}{\lambda+1}, \frac{4\,c_1\,k_{\lambda}}{\lambda+1}, -\frac{2\,c_1\left(\left(\lambda^2+2\,\lambda\right)k_{\lambda}+k(\lambda)\,(\lambda+1)\right)}{\lambda+1}, -\frac{2\,c_1\left(k_{\lambda}+k(\lambda)\,(\lambda+1)\right)}{\lambda+1}\right\} \tag{92}$$

```
E > rifsimp({case1_varconds[],diff(k(lambda),lambda)<>0},casesplit):%
    [Solved]:var1:=%;
```

$$var1 := [c_1 = 0] \tag{93}$$

```
E >
E >
E >
E > ### CASE 3
E > subs(soln3,P_u*eqn_u+P_v*eqn_v+P_w*eqn_w):dsubs(k_soln3,%)
    :char_eqn3:=simplify(%):
E > EulerLagrange(char_eqn3):simplify(%,size):algsubs(solve(strain,{w
    [2]^2})[],expand(%)):simplify(%) assuming(lambda>0):convert(%,
    diff):remove(has,indets(%,name),{lambda,soln3Cs[],p,k0}):{seq
    (coeffs(expand(%%[i]),%),i=1..nops(%%))}:subs((lambda+1)^p=alpha,
    %):simplify(%):case3_varconds:=%;
```

Warning, solving for expressions other than names or functions is not recommended.

$$case3\_varconds := \left\{\frac{\left(-c_1+3\,c_3\right)p-4\,c_1+4\,c_3}{p}, \frac{\alpha\,k0\left((p+4)\,c_1+(-3\,p-4)\,c_3\right)}{(\lambda+1)^2}, \right. \tag{94}$$

$$\frac{\alpha\, k0\left(\left(c_1-3c_3\right)p+4c_1-4c_3\right)}{p}, \frac{\alpha\, k0\left(\left(c_1-3c_3\right)p+4c_1-4c_3\right)\left(p+\left(\lambda+1\right)^2\right)}{p\left(\lambda+1\right)^2},$$

$$\frac{\alpha\, k0\left(\left(c_1-3c_3\right)p+4c_1-4c_3\right)\left(1+\left(p+1\right)\lambda^2+\left(2p+2\right)\lambda\right)}{p\left(\lambda+1\right)^2},$$

$$-\frac{2\alpha\, k0\left(\left(p+4\right)c_1+\left(-3p-4\right)c_3\right)}{\left(\lambda+1\right)^2}, -\frac{\alpha\, k0\left(\left(p+4\right)c_1+\left(-3p-4\right)c_3\right)}{\left(\lambda+1\right)^2},$$

$$\left.\frac{2\alpha\, k0\left(\left(p+4\right)c_1+\left(-3p-4\right)c_3\right)}{\left(\lambda+1\right)^2}\right\}$$

```
E > rifsimp({case3_varconds[],sys3_p[]},arbitrary={lambda,alpha,k0},
    casesplit):simplify(%[Solved]):var3:=%;
```

$$var3 := \left[c_1 = \frac{c_3\left(3p+4\right)}{p+4}\right] \tag{95}$$

```
E >
E >
E >
E > ### CASE 5
E > subs(soln5,P_u*eqn_u+P_v*eqn_v+P_w*eqn_w):dsubs(k_soln5,%)
    :char_eqn5:=simplify(%):
E > EulerLagrange(%):simplify(%,size):algsubs(solve(strain,{w[2]^2})
    [],expand(%)):simplify(%) assuming(lambda>0):convert(%,diff)
    :remove(has,indets(%,name),{lambda,soln5Cs[],k0}):{seq(coeffs
    (expand(%%[i]),%),i=1..nops(%%))}:simplify(%):case5_varconds:=%;
```

Warning, solving for expressions other than names or functions is not recommended.

$$case5\_varconds := \left\{2c_4, -\frac{16c_4 k0}{\left(\lambda+1\right)^6}, -\frac{8c_4 k0}{\left(\lambda+1\right)^6}, \frac{8c_4 k0}{\left(\lambda+1\right)^6}, \frac{16c_4 k0}{\left(\lambda+1\right)^6}, -\frac{2k0\, c_4}{\left(\lambda+1\right)^4},\right. \tag{96}$$

$$\left.\frac{6\left(\lambda^2+2\lambda-\frac{1}{3}\right)k0\, c_4}{\left(\lambda+1\right)^6}, -\frac{2k0\, c_4\left(\lambda+3\right)\left(\lambda-1\right)}{\left(\lambda+1\right)^6}\right\}$$

```
E > rifsimp({case5_varconds[]},arbitrary={lambda,k0},casesplit)
    :simplify(%[Solved]):var5:=%;
```

$$var5 := \left[c_4 = 0\right] \tag{97}$$

```
E >
E >
E >
E > ##### Derive T,X in the conservation laws
```

```
E >
E > ### CASE 1
E >
E > ibp1:=subs(var1,char_eqn1):EulerLagrange(ibp1):simplify(%,size);
```
$$[0, 0, 0] \tag{98}$$
```
E >
E > ibp1:indets(%,name) minus soln1Cs;
```
$$\{t, x, u[\,], u_1, u_2, u_{1,1}, u_{2,2}, v[\,], v_1, v_2, v_{1,1}, v_{2,2}, w[\,], w_1, w_2, w_{1,1}, w_{2,2}\} \tag{99}$$
```
E > coeff(ibp1,u[1,1]):Tterm1:=int(%,u[1]):
E > ibp1-TotalDiff(Tterm1,t):ibp2:=simplify(%):indets(%,name) minus
    soln1Cs;
```
$$\{t, x, u[\,], u_1, u_2, u_{1,2}, u_{2,2}, v[\,], v_1, v_2, v_{1,1}, v_{2,2}, w[\,], w_1, w_2, w_{1,1}, w_{2,2}\} \tag{100}$$
```
E > coeff(ibp2,v[1,1]):Tterm2:=int(%,v[1]):
E > ibp2-TotalDiff(Tterm2,t):ibp3:=simplify(%):indets(%,name) minus
    soln1Cs;
```
$$\{t, x, u[\,], u_1, u_2, u_{1,2}, u_{2,2}, v[\,], v_1, v_2, v_{1,2}, v_{2,2}, w[\,], w_1, w_2, w_{1,1}, w_{2,2}\} \tag{101}$$
```
E > coeff(ibp3,w[1,1]):Tterm3:=int(%,w[1]):
E > ibp3-TotalDiff(Tterm3,t):ibp4:=simplify(%):indets(%,name) minus
    soln1Cs;
```
$$\{t, x, u[\,], u_1, u_2, u_{1,2}, u_{2,2}, v[\,], v_1, v_2, v_{1,2}, v_{2,2}, w[\,], w_1, w_2, w_{1,2}, w_{2,2}\} \tag{102}$$
```
E > coeff(ibp4,u[2,2]):collect(%,indets(%,function));#Xterm1:=Expand
    (int(%,u[2]));
```
$$\begin{aligned}&\left(c_{12} v[\,] + c_{16} u_1 + c_2 u_2 - c_3 t + c_5 w[\,] - c_{14}\right) k\left(\sqrt{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1} - 1\right) \\ &\quad - \frac{1}{\sqrt{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}} \Big(\left(1 + u_2\right) \Big(-v_2^2 c_2 + \Big(-c_{16} v_1 + \left(x + u[\,]\right) c_{12} + c_{18} t \\ &\quad - c_6 w[\,] + c_{13}\Big) v_2 - w_2^2 c_2 + \left(-c_{16} w_1 + \left(x + u[\,]\right) c_5 + c_4 t + c_6 v[\,] + c_{15}\right) w_2 + \left(1 + u_2\right) \Big(\\ &\quad -c_{12} v[\,] - c_{16} u_1 - c_2 u_2 + c_3 t - c_5 w[\,] + c_{14}\Big)\Big) \mathrm{D}(k)\left(\sqrt{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1} - 1\right)\Big)\end{aligned} \tag{103}$$
```
E > #ibp3-TotalDiff(Xterm1,x):ibp4:=simplify(%):indets(%,name) minus
    soln1Cs;
E >
E > soln1Cs minus indets(var1,name):conslaw1Cs:=%;
```
$$conslaw1Cs := \{c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_{18}, c_2, c_3, c_4, c_5, c_6\} \tag{104}$$
```
E >
E >
E >
E > ### CASE 3
E >
```

```
E > ibp1:=subs(var3,char_eqn3):EulerLagrange(ibp1):simplify(%,size);
```
$$[0, 0, 0] \tag{105}$$

```
E >
E > ibp1:indets(%,name) minus soln3Cs;
```
$$\{k0, p, t, x, u[\,], u_1, u_2, u_{1,1}, u_{2,2}, v[\,], v_1, v_2, v_{1,1}, v_{2,2}, w[\,], w_1, w_2, w_{1,1}, w_{2,2}\} \tag{106}$$

```
E > coeff(ibp1,u[1,1]):Tterm1:=int(%,u[1]):
E > ibp1-TotalDiff(Tterm1,t):ibp2:=simplify(%):indets(%,name) minus
    soln3Cs;
```
$$\{k0, p, t, x, u[\,], u_1, u_2, u_{1,2}, u_{2,2}, v[\,], v_1, v_2, v_{1,1}, v_{2,2}, w[\,], w_1, w_2, w_{1,1}, w_{2,2}\} \tag{107}$$

```
E > coeff(ibp2,v[1,1]):Tterm2:=int(%,v[1]):
E > ibp2-TotalDiff(Tterm2,t):ibp3:=simplify(%):indets(%,name) minus
    soln3Cs;
```
$$\{k0, p, t, x, u[\,], u_1, u_2, u_{1,2}, u_{2,2}, v[\,], v_1, v_2, v_{1,2}, v_{2,2}, w[\,], w_1, w_2, w_{1,1}, w_{2,2}\} \tag{108}$$

```
E > coeff(ibp3,w[1,1]):Tterm3:=int(%,w[1]):
E > ibp3-TotalDiff(Tterm3,t):ibp4:=simplify(%):indets(%,name) minus
    soln3Cs;
```
$$\{k0, p, t, x, u[\,], u_1, u_2, u_{1,2}, u_{2,2}, v[\,], v_1, v_2, v_{1,2}, v_{2,2}, w[\,], w_1, w_2, w_{1,2}, w_{2,2}\} \tag{109}$$

```
E > coeff(ibp4,u[2,2]):Xterm1:=int(%,u[2]):
E > ibp4-TotalDiff(Xterm1,x):simplify(%):ibp5:=collect(%,[u[2,2],v[2,
    2],w[2,2]],simplify):indets(%,name) minus soln3Cs;
```
$$\{k0, p, t, x, u_1, u_2, u_{1,2}, v_1, v_2, v_{1,2}, w_1, w_2, w_{1,2}\} \tag{110}$$

```
E > coeff(ibp5,u[1,2]):coeff(%,u[1],0):Tterm4:=int(%,u[2]):
E > ibp5-TotalDiff(Tterm4,t):simplify(%):ibp6:=collect(%,[u[2,2],v[2,
    2],w[2,2]],simplify):indets(%,name) minus soln3Cs;
```
$$\{x, u_1, u_{1,2}, v_1, v_{1,2}, w_1, w_{1,2}\} \tag{111}$$

```
E > coeff(ibp6,u[1,2]):Xterm2:=int(%,u[1]):
E > ibp6-TotalDiff(Xterm2,x):ibp7:=simplify(%):indets(%,name) minus
    soln3Cs;
```
$$\{x, u_1, v_1, v_{1,2}, w_1, w_{1,2}\} \tag{112}$$

```
E > coeff(ibp7,v[1,2]):Xterm3:=int(%,v[1]):
E > ibp7-TotalDiff(Xterm3,x):ibp8:=simplify(%):indets(%,name) minus
    soln3Cs;
```
$$\{x, u_1, v_1, w_1, w_{1,2}\} \tag{113}$$

```
E > coeff(ibp8,w[1,2]):Xterm4:=int(%,w[1]):
E > ibp8-TotalDiff(Xterm4,x):ibp9:=simplify(%):indets(%,name) minus
    soln3Cs;
```
$$\{u_1, v_1, w_1\} \tag{114}$$

```
E > coeff(ibp9,u[1]):Tterm5:=int(%,u[]):
E > ibp9-TotalDiff(Tterm5,t):ibp10:=simplify(%):indets(%,name) minus
    soln3Cs;
```
$$\{v_1, w_1\} \tag{115}$$

```
E > coeff(ibp10,v[1]):Tterm6:=int(%,v[]):
E > ibp10-TotalDiff(Tterm6,t):ibp11:=simplify(%):indets(%,name) minus
    soln3Cs;
```

$$\{w_1\} \tag{116}$$

```
E > coeff(ibp11,w[1]):Tterm7:=int(%,w[]):
E > ibp11-TotalDiff(Tterm7,t):ibp12:=simplify(%):indets(%,name) minus
    soln3Cs;
```

$$\varnothing \tag{117}$$

```
E >
E >
E > Tterm1+Tterm2+Tterm3+Tterm4+Tterm5+Tterm6+Tterm7:T3:=simplify(%):
E > Xterm1+Xterm2+Xterm3+Xterm4:X3:=simplify(%):
E > TotalDiff(T3,t)+TotalDiff(X3,x)-ibp1:simplify(%):collect(%,[u[2,
    2],v[2,2],w[2,2]],simplify):check3:=simplify(%);
```

$$check3 := 0 \tag{118}$$

```
E >
E > soln3Cs minus indets(lhs(var3[]),name):conslaw3Cs:=%;
```

$$conslaw3Cs := \{\_C24, c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_{18}, c_2, c_3, c_4, c_5, c_6\} \tag{119}$$

```
E >
E >
E > collect(T3,[conslaw3Cs[],u[1],v[1],w[1],u[2],v[2],w[2]],simplify)
    :T3collect:=%;
```

$$T3collect := \left(t\,u_1 - u[\,]\right) c_{18} + \left(t\,v_1 - v[\,]\right) c_{12} + \left(-u_1\,v[\,] + \left(x + u[\,]\right) v_1\right) c_{13} + \left(t\,w_1 - w[\,]\right) c_{14} + c_{15}\,v_1 + c_{16}\,w_1 + \left(-\frac{u_1^2}{2} - \frac{v_1^2}{2} - \frac{w_1^2}{2} - \frac{k0\left(u_2^2 + v_2^2 + w_2^2 + 2\,u_2 + 1\right)^{1+\frac{p}{2}}}{p+2}\right) c_2 + \_C24\,u_1 + \left(-\frac{t\left(3\,p+4\right) u_1^2}{2\,p+8} + \left(-x\,u_2 + \frac{p\,u[\,] - 4\,x}{p+4}\right) u_1 - \frac{t\left(3\,p+4\right) v_1^2}{2\,p+8} + \left(-v_2\,x + \frac{p\,v[\,]}{p+4}\right) v_1 - \frac{t\left(3\,p+4\right) w_1^2}{2\,p+8} + \left(-w_2\,x + \frac{p\,w[\,]}{p+4}\right) w_1 - \frac{k0\left(u_2^2 + v_2^2 + w_2^2 + 2\,u_2 + 1\right)^{1+\frac{p}{2}}\left(3\,p+4\right) t}{\left(p+4\right)\left(p+2\right)}\right) c_3 + \left(-u_2\,u_1 - v_1\,v_2 - w_1\,w_2\right) c_4 + \left(-u_1\,w[\,] + \left(x + u[\,]\right) w_1\right) c_5 + \left(v[\,]\,w_1 - w[\,]\,v_1\right) c_6 \tag{120}$$

```
E > collect(X3,[conslaw3Cs[],k0,u[1],v[1],w[1]],simplify):X3collect:=
```

%;

$$X3collect := -\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} t\left(1+u_2\right) k0\, c_{18} - \left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} c_{12}\, k0\, t\, v_2 - \left(-v[\,]\, u_2 + \left(x+u[\,]\right) v_2 - v[\,]\right)\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} k0\, c_{13} - \left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} c_{14}\, k0\, t\, w_2 - \left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} c_{15}\, k0\, v_2 - \left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} c_{16}\, k0\, w_2 + \left(\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}\left(1+u_2\right) u_1 + \left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} v_2 v_1 + \left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} w_2 w_1\right) k0\, c_2 - \left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}\left(1+u_2\right) k0\_C24 + \left(\left(\frac{3\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}\left(1+u_2\right) t\left(p+\frac{4}{3}\right) u_1}{p+4} + \frac{\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}\left(3p+4\right) t\, v_2 v_1}{p+4} + \frac{\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}\left(3p+4\right) t\, w_2 w_1}{p+4} + \frac{1}{(p+4)(p+2)}\left(\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}\left(\left(\left(u_2^2+v_2^2+w_2^2+u_2\right) x - u[\,]\, u_2 - v[\,]\, v_2 - w[\,]\, w_2 - u[\,]\right) p^2 + \left(\left(5u_2^2+5v_2^2+5w_2^2+8u_2+3\right) x - 2u[\,]\, u_2 - 2v[\,]\, v_2 - 2w[\,]\, w_2 - 2u[\,]\right) p + 4x\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)\right)\right)\right) k0 + \frac{x u_1^2}{2} + \frac{x v_1^2}{2} + \frac{w_1^2 x}{2}\right) c_3 \tag{121}$$

$$+ \left(\frac{\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}\left(\left(u_2^2+v_2^2+w_2^2+u_2\right) p + u_2^2+v_2^2+w_2^2-1\right) k0}{p+2} + \frac{u_1^2}{2}\right.$$

$$+\frac{v_1^2}{2}+\frac{w_1^2}{2}\Bigg)c_4-\left(-w[\,]u_2+(x+u[\,])w_2-w[\,]\right)\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}k0\,c_5$$

$$-\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}\left(w_2v[\,]-v_2w[\,]\right)k0\,c_6$$

E >

E > # densities

```
E > seq(print(conslaw3Cs[i], coeff(T3collect,conslaw3Cs[i])),i=1..
    nops(conslaw3Cs));
```

$$c_{18},\, t\,u_1-u[\,]$$

$$c_{12},\, t\,v_1-v[\,]$$

$$c_{13},\, -u_1v[\,]+(x+u[\,])\,v_1$$

$$c_{14},\, t\,w_1-w[\,]$$

$$c_{15},\, v_1$$

$$c_{16},\, w_1$$

$$c_2,\, -\frac{u_1^2}{2}-\frac{v_1^2}{2}-\frac{w_1^2}{2}-\frac{k0\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{1+\frac{p}{2}}}{p+2}$$

$$\_C24,\, u_1$$

$$c_3,\, -\frac{t\,(3p+4)\,u_1^2}{2p+8}+\left(-x\,u_2+\frac{p\,u[\,]-4x}{p+4}\right)u_1-\frac{t\,(3p+4)\,v_1^2}{2p+8}+\left(-v_2x+\frac{p\,v[\,]}{p+4}\right)v_1$$

$$-\frac{t\,(3p+4)\,w_1^2}{2p+8}+\left(-w_2x+\frac{p\,w[\,]}{p+4}\right)w_1-\frac{k0\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{1+\frac{p}{2}}(3p+4)\,t}{(p+4)\,(p+2)}$$

$$c_4,\, -u_2u_1-v_1v_2-w_1w_2$$

$$c_5,\, -u_1w[\,]+(x+u[\,])\,w_1$$

$$c_6,\, v[\,]\,w_1-v_1w[\,] \tag{122}$$

E >

E > # fluxes

```
E > seq(print(conslaw3Cs[i], coeff(X3collect,conslaw3Cs[i])),i=1..
    nops(conslaw3Cs));
```

$$c_{18},\, -\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}t\,(1+u_2)\,k0$$

$$c_{12},\, -\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}k0\,t\,v_2$$

$$c_{13},\, -\left(-v[\,]u_2+(x+u[\,])\,v_2-v[\,]\right)\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}k0$$

$$c_{14}, -\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} k0\, t\, w_2$$

$$c_{15}, -\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} k0\, v_2$$

$$c_{16}, -\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} k0\, w_2$$

$$c_2, \left(\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}(1+u_2)\,u_1+\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} v_2 v_1+\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} w_2 w_1\right) k0$$

$$\_C24, -\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}(1+u_2)\,k0$$

$$c_3, \left(\frac{3\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}(1+u_2)\,t\left(p+\frac{4}{3}\right)u_1}{p+4}+\frac{\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}(3p+4)\,t\,v_2 v_1}{p+4}+\frac{\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}(3p+4)\,t\,w_2 w_1}{p+4}+\frac{1}{(p+4)(p+2)}\left(\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}\left(\left(\left(u_2^2+v_2^2+w_2^2+u_2\right)x-u[\,]u_2-v[\,]v_2-w[\,]w_2-u[\,]\right)p^2+\left(\left(5u_2^2+5v_2^2+5w_2^2+8u_2+3\right)x-2u[\,]u_2-2v[\,]v_2-2w[\,]w_2-2u[\,]\right)p+4x\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)\right)\right)k0+\frac{x u_1^2}{2}+\frac{x v_1^2}{2}+\frac{w_1^2 x}{2}\right.$$

$$c_4, \frac{\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}\left(\left(u_2^2+v_2^2+w_2^2+u_2\right)p+u_2^2+v_2^2+w_2^2-1\right)k0}{p+2}+\frac{u_1^2}{2}+\frac{v_1^2}{2}+\frac{w_1^2}{2}$$

$$c_5, -\left(-w[\,]u_2+(x+u[\,])\,w_2-w[\,]\right)\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}} k0$$

$$c_6, -\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^{\frac{p}{2}}\left(w_2 v[\,]-v_2 w[\,]\right)k0 \tag{123}$$

E >

```
E >
E > ### CASE 5
E >
E > ibp1:=subs(var5,char_eqn5):EulerLagrange(ibp1):simplify(%,size);
```

$$[0, 0, 0] \tag{124}$$

```
E >
E > ibp1:indets(%,name) minus soln5Cs;
```

$$\{k0, t, x, u[\ ], u_1, u_2, u_{1,1}, u_{2,2}, v[\ ], v_1, v_2, v_{1,1}, v_{2,2}, w[\ ], w_1, w_2, w_{1,1}, w_{2,2}\} \tag{125}$$

```
E > coeff(ibp1,u[1,1]):Tterm1:=int(%,u[1]):
E > ibp1-TotalDiff(Tterm1,t):ibp2:=simplify(%):indets(%,name) minus
    soln5Cs;
```

$$\{k0, t, x, u[\ ], u_1, u_2, u_{1,2}, u_{2,2}, v[\ ], v_1, v_2, v_{1,1}, v_{2,2}, w[\ ], w_1, w_2, w_{1,1}, w_{2,2}\} \tag{126}$$

```
E > coeff(ibp2,v[1,1]):Tterm2:=int(%,v[1]):
E > ibp2-TotalDiff(Tterm2,t):ibp3:=simplify(%):indets(%,name) minus
    soln5Cs;
```

$$\{k0, t, x, u[\ ], u_1, u_2, u_{1,2}, u_{2,2}, v[\ ], v_1, v_2, v_{1,2}, v_{2,2}, w[\ ], w_1, w_2, w_{1,1}, w_{2,2}\} \tag{127}$$

```
E > coeff(ibp3,w[1,1]):Tterm3:=int(%,w[1]):
E > ibp3-TotalDiff(Tterm3,t):ibp4:=simplify(%):indets(%,name) minus
    soln5Cs;
```

$$\{k0, t, x, u[\ ], u_1, u_2, u_{1,2}, u_{2,2}, v[\ ], v_1, v_2, v_{1,2}, v_{2,2}, w[\ ], w_1, w_2, w_{1,2}, w_{2,2}\} \tag{128}$$

```
E > coeff(ibp4,u[2,2]):Xterm1:=int(%,u[2]):
E > ibp4-TotalDiff(Xterm1,x):simplify(%):ibp5:=collect(%,[u[2,2],v[2,
    2],w[2,2]],simplify):indets(%,name) minus soln5Cs;
```

$$\{k0, t, x, u[\ ], u_1, u_2, u_{1,2}, v[\ ], v_1, v_2, v_{1,2}, w[\ ], w_1, w_2, w_{1,2}\} \tag{129}$$

```
E > coeff(ibp5,u[1,2]):coeff(%,u[1],0):Tterm4:=int(%,u[2]):
E > ibp5-TotalDiff(Tterm4,t):simplify(%):ibp6:=collect(%,[u[2,2],v[2,
    2],w[2,2]],simplify):indets(%,name) minus soln5Cs;
```

$$\{x, u[\ ], u_1, u_{1,2}, v[\ ], v_1, v_{1,2}, w[\ ], w_1, w_{1,2}\} \tag{130}$$

```
E > coeff(ibp6,u[1,2]):Xterm2:=int(%,u[1]):
E > ibp6-TotalDiff(Xterm2,x):ibp7:=simplify(%):indets(%,name) minus
    soln5Cs;
```

$$\{x, u[\ ], u_1, v[\ ], v_1, v_{1,2}, w[\ ], w_1, w_{1,2}\} \tag{131}$$

```
E > coeff(ibp7,v[1,2]):Xterm3:=int(%,v[1]):
E > ibp7-TotalDiff(Xterm3,x):ibp8:=simplify(%):indets(%,name) minus
    soln5Cs;
```

$$\{x, u[\ ], u_1, v[\ ], v_1, w[\ ], w_1, w_{1,2}\} \tag{132}$$

```
E > coeff(ibp8,w[1,2]):Xterm4:=int(%,w[1]):
E > ibp8-TotalDiff(Xterm4,x):ibp9:=simplify(%):indets(%,name) minus
    soln5Cs;
```

$$\{x, u[\ ], u_1, v[\ ], v_1, w[\ ], w_1\} \tag{133}$$

```
E > coeff(ibp9,u[1]):Tterm5:=int(%,u[]):
E > ibp9-TotalDiff(Tterm5,t):ibp10:=simplify(%):indets(%,name) minus
    soln5Cs;
```

$$\{v[\,],v_1,w[\,],w_1\} \tag{134}$$

```
E > coeff(ibp10,v[1]):Tterm6:=int(%,v[]):
E > ibp10-TotalDiff(Tterm6,t):ibp11:=simplify(%):indets(%,name) minus
    soln5Cs;
```

$$\{w[\,],w_1\} \tag{135}$$

```
E > coeff(ibp11,w[1]):Tterm7:=int(%,w[]):
E > ibp11-TotalDiff(Tterm7,t):ibp12:=simplify(%):indets(%,name) minus
    soln5Cs;
```

$$\varnothing \tag{136}$$

```
E >
E >
E > Tterm1+Tterm2+Tterm3+Tterm4+Tterm5+Tterm6+Tterm7:T5:=simplify(%):
E > Xterm1+Xterm2+Xterm3+Xterm4:X5:=simplify(%):
E > TotalDiff(T5,t)+TotalDiff(X5,x)-ibp1:simplify(%):collect(%,[u[2,
    2],v[2,2],w[2,2]],simplify):check5:=simplify(%);
```

$$check5 := 0 \tag{137}$$

```
E >
E > soln5Cs minus indets(var5,name):conslaw5Cs:=%;
```

$$conslaw5Cs := \{c_1, c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_{18}, c_{19}, c_2, c_{20}, c_{21}, c_3, c_5\} \tag{138}$$

```
E >
E >
E > collect(T5,[conslaw5Cs[],u[1],v[1],w[1],u[2],v[2],w[2]],simplify)
    :T5collect:=%;
```

$$T5collect := \left(-\frac{u_1^2 t^2}{4} + \frac{t\,(x+u[\,])\,u_1}{2} - \frac{w_1^2 t^2}{4} - \frac{x\,u[\,]}{2} - \frac{w[\,]^2}{4} + \frac{t^2 k0}{4u_2^2+4v_2^2+4w_2^2+8u_2+4} - \frac{v[\,]^2}{4} - \frac{u[\,]^2}{4} - \frac{v_1^2 t^2}{4} + \frac{t\,w[\,]\,w_1}{2} + \frac{t\,v[\,]\,v_1}{2}\right) c_1 + \Big(-u_1 v[\,] + (x+u[\,])\,v_1\Big)\,c_{18} + \Big(-u_1 w[\,] + (x+u[\,])\,w_1\Big)\,c_{12} + \Big(v[\,]\,w_1 - w[\,]\,v_1\Big)\,c_{13} + c_{14} u_1 + c_{15} v_1 + c_{16} w_1 + \Big(t\,u_1 - u[\,]\Big)\,c_{19} + \Big(t\,w_1 - w[\,]\Big)\,c_{20} + \Big(t\,v_1 - v[\,]\Big)\,c_{21} + \Bigg(-\frac{t\,u_1^2}{2} + \left(\frac{u[\,]}{2} + \frac{x}{2}\right) u_1 + \frac{v[\,]\,v_1}{2} - \frac{t\,w_1^2}{2} - \frac{t\,v_1^2}{2} + \frac{2\,t\,k0}{4u_2^2+4v_2^2+4w_2^2+8u_2+4} \tag{139}$$

$$+\frac{w[\,]\,w_1}{2}\Bigg)c_2+\left(-\frac{v_1^2}{2}-\frac{u_1^2}{2}+\frac{2\,k0}{4\,u_2^2+4\,v_2^2+4\,w_2^2+8\,u_2+4}-\frac{w_1^2}{2}\right)c_3+\Big(-u_1\,u_2$$

$$-v_1\,v_2-w_1\,w_2\Big)\,c_5$$

```
E > collect(X5,[conslaw5Cs[],k0,u[1],v[1],w[1]],simplify):X5collect:=
    %;
```

$$X5collect := \left(\frac{t^2\left(1+u_2\right)u_1}{2\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}+\frac{t^2\,v_2\,v_1}{2\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}\right. \tag{140}$$

$$\left.+\frac{t^2\,w_2\,w_1}{2\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}-\frac{t\left(\left(x+u[\,]\right)u_2+v[\,]\,v_2+w[\,]\,w_2+x+u[\,]\right)}{2\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}\right)k0\,c_1$$

$$+\frac{\left(v[\,]\,u_2+\left(-x-u[\,]\right)v_2+v[\,]\right)k0\,c_{18}}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}+\frac{\left(w[\,]\,u_2+\left(-x-u[\,]\right)w_2+w[\,]\right)k0\,c_{12}}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}$$

$$+\frac{\left(-w_2\,v[\,]+v_2\,w[\,]\right)k0\,c_{13}}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}+\frac{\left(-u_2-1\right)k0\,c_{14}}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}$$

$$-\frac{v_2\,k0\,c_{15}}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}-\frac{w_2\,k0\,c_{16}}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}$$

$$-\frac{t\left(1+u_2\right)k0\,c_{19}}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}-\frac{k0\,t\,w_2\,c_{20}}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}$$

$$-\frac{k0\,t\,v_2\,c_{21}}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}+\left(\frac{t\left(1+u_2\right)u_1}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}\right.$$

$$+\frac{t\,v_2\,v_1}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}+\frac{t\,w_2\,w_1}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}$$

$$\left.+\frac{\left(-x-u[\,]\right)u_2-v[\,]\,v_2-w[\,]\,w_2-x-u[\,]}{2\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}\right)k0\,c_2+\left(\frac{\left(1+u_2\right)u_1}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}\right.$$

$$\left.+\frac{v_2\,v_1}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}+\frac{w_1\,w_2}{\left(u_2^2+v_2^2+w_2^2+2\,u_2+1\right)^2}\right)k0\,c_3$$

$$+\left(\frac{\left(3u_2^2+3v_2^2+3w_2^2+4u_2+1\right)k0}{2\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^2}+\frac{u_1^2}{2}+\frac{v_1^2}{2}+\frac{w_1^2}{2}\right)c_5$$

```
E >
E > # densities
E > seq(print(conslaw5Cs[i], coeff(T5collect,conslaw5Cs[i])),i=1..
    nops(conslaw5Cs));
```

$$c_1, -\frac{u_1^2t^2}{4}+\frac{t(x+u[\,])u_1}{2}-\frac{w_1^2t^2}{4}-\frac{xu[\,]}{2}-\frac{w[\,]^2}{4}+\frac{t^2k0}{4u_2^2+4v_2^2+4w_2^2+8u_2+4}-\frac{v[\,]^2}{4}-\frac{u[\,]^2}{4}-\frac{v_1^2t^2}{4}+\frac{tw[\,]w_1}{2}+\frac{tv[\,]v_1}{2}$$

$$c_{18}, -u_1v[\,]+(x+u[\,])v_1$$

$$c_{12}, -u_1w[\,]+(x+u[\,])w_1$$

$$c_{13}, v[\,]w_1-w[\,]v_1$$

$$c_{14}, u_1$$

$$c_{15}, v_1$$

$$c_{16}, w_1$$

$$c_{19}, tu_1-u[\,]$$

$$c_{20}, tw_1-w[\,]$$

$$c_{21}, tv_1-v[\,]$$

$$c_2, -\frac{tu_1^2}{2}+\left(\frac{u[\,]}{2}+\frac{x}{2}\right)u_1+\frac{v[\,]v_1}{2}-\frac{tw_1^2}{2}-\frac{tv_1^2}{2}+\frac{2tk0}{4u_2^2+4v_2^2+4w_2^2+8u_2+4}+\frac{w[\,]w_1}{2}$$

$$c_3, -\frac{v_1^2}{2}-\frac{u_1^2}{2}+\frac{2k0}{4u_2^2+4v_2^2+4w_2^2+8u_2+4}-\frac{w_1^2}{2}$$

$$c_5, -u_1u_2-v_1v_2-w_1w_2 \tag{141}$$

```
E >
E > # fluxes
E > seq(print(conslaw5Cs[i], coeff(X5collect,conslaw5Cs[i])),i=1..
    nops(conslaw5Cs));
```

$$c_1, \left(\frac{t^2(1+u_2)u_1}{2\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^2}+\frac{t^2v_2v_1}{2\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^2}+\frac{t^2w_2w_1}{2\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^2}-\frac{t\left((x+u[\,])u_2+v[\,]v_2+w[\,]w_2+x+u[\,]\right)}{2\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^2}\right)k0$$

$$c_{18}, \frac{\left(v[\,]u_2+(-x-u[\,])v_2+v[\,]\right)k0}{\left(u_2^2+v_2^2+w_2^2+2u_2+1\right)^2}$$

$$c_{12}, \frac{\left(w[\,]\, u_2 + \left(-x - u[\,]\right) w_2 + w[\,]\right) k0}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2}$$

$$c_{13}, \frac{\left(-w_2\, v[\,] + v_2\, w[\,]\right) k0}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2}$$

$$c_{14}, \frac{\left(-u_2 - 1\right) k0}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2}$$

$$c_{15}, -\frac{v_2\, k0}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2}$$

$$c_{16}, -\frac{w_2\, k0}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2}$$

$$c_{19}, -\frac{t\left(1 + u_2\right) k0}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2}$$

$$c_{20}, -\frac{k0\, t\, w_2}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2}$$

$$c_{21}, -\frac{k0\, t\, v_2}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2}$$

$$c_2, \left(\frac{t\left(1 + u_2\right) u_1}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2} + \frac{t\, v_2\, v_1}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2} + \frac{t\, w_2\, w_1}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2} + \frac{\left(-x - u[\,]\right) u_2 - v[\,]\, v_2 - w[\,]\, w_2 - x - u[\,]}{2\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2}\right) k0$$

$$c_3, \left(\frac{\left(1 + u_2\right) u_1}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2} + \frac{v_2\, v_1}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2} + \frac{w_1\, w_2}{\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2}\right) k0$$

$$c_5, \frac{\left(3\, u_2^2 + 3\, v_2^2 + 3\, w_2^2 + 4\, u_2 + 1\right) k0}{2\left(u_2^2 + v_2^2 + w_2^2 + 2\, u_2 + 1\right)^2} + \frac{u_1^2}{2} + \frac{v_1^2}{2} + \frac{w_1^2}{2} \tag{142}$$

E >

E >